



Blockchain and Big Data Analytics in Agriculture: A Review of Digital Innovations

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This work was carried out in collaboration among all authors. All authors read and approved the final manuscript.

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ABSTRACT

In order to solve difficult issues in food and agriculture, information systems and big data analytics (BDA) are becoming more and more important. A lot of big data is used in retail and food distribution, but not enough in production, decision-making, or durability. In order to combine various data types, including sensor data, satellite images, and weather forecasts, BDA uses cutting-edge methods like machine learning and artificial intelligence. It then seeks out actionable data that can be applied to enhance agriculture, food processing, and preservation. In order to fully realize the potential of big data in transforming the future of food, the review suggested that future research should focus on interdisciplinary collaboration, addressing ethical considerations, and developing robust data governance frameworks – the ethical and regulatory frameworks need to keep up with technological advancements. This systematic review outlines recent advancements and emphasizes trends, challenges, and aspects that require further investigation regarding the use of BDA in the agri-food industry. It emphasizes how BDA can enhance quality assurance, precision farming, supply chain efficiency, and sustainable food systems. The primary challenges involve the sharing of data across systems, infrastructure problems, privacy issues, and differences in the pace at which stakeholders embrace technologies.

Keywords: *Traceability; agriculture supply chain; agri-food industry; food processing; data-driven preservation; Artificial Intelligence.*

1. INTRODUCTION

Computers can mimic parts of human thought and behavior through artificial intelligence (AI), a collection of technologies that mimic human intelligence and enable autonomous learning, reasoning, planning, and decision-making. Deep learning, computer vision, natural language processing, machine learning (ML) and other technologies form the foundation of artificial intelligence and can be used in a wide range of domains and sectors. "Big data" is the term used to describe the vast and varied collection of data (Akoka et al., 2017) that is usually produced by numerous sensors or mobile devices, scraped from the internet and other sources, and comprises both organized and unstructured data, including text, photos, and videos. These data sets must be stored, processed, and analyzed using specialized technologies often commonly referred to as "big data technologies" because they are typically characterized by high speed, high density, and high dimensionality. Big data analytics may assist organizations in the food industry to anticipate market demand, optimize the supply chain, enhance food safety and quality, and gain more opportunities and competitive advantages (Schroeder, 2016). As the two crucial technologies that work well together, artificial intelligence and big data are crucial to the food industry and are having a large influence on innovation and development (Misra et al., 2020).

AI is currently being used extensively in manufacturing, healthcare (Sahu et al., 2022), finance, transportation, gaming, and processing (Ramakrishna et al., 2020; Serazetdinova et al., 2019). In these sectors, the use of AI can boost productivity, cut expenses, and enhance service quality.

"For the future well-being of the global society, agricultural innovation is a necessity. As it has been throughout history, having access to adequate amounts of food is an uncertainty for many individuals across the world. For others, availability of reasonably safe, affordable food is somewhat taken for granted. For both groups, simply continuing the practices of our current agricultural and food system is not sufficient nor tenable as we look to the future" (Sonka, 2021). "Four stressors are particularly significant among the numerous others that affect that system. Firstly It is anticipated that population expansion will continue worldwide, especially in regions with comparatively low levels of food security. Secondly this challenge suggests the need for the environmental implications of agricultural production are increasingly acknowledged to have both short-term and long-term unfavorable repercussions, notwithstanding the need to produce more food. Thirdly the prospects of a changing, more variable climate contribute to the need to enhance the resilience of current agricultural practices" (Sonka,

2021). Fourth, consumers are calling for an even safer and more nutrient-dense food supply in both developed and developing nations.

According to one viewpoint, these difficulties are overwhelming and could appear insurmountable. However, this environment is not new to humanity and is more the rule than the exception from a historical standpoint. Innovation in agriculture is crucial to the survival and advancement of society. Indeed, distinguished academics using advanced mathematical models boldly forecasted fifty years ago that by 10 years, the world would experience widespread starvation as a result of ongoing food shortages (Meadows et al., 1972). However, societal shifts and agricultural innovation stepped up to the plate and helped lower the global number of starving people. These days, agricultural production systems are developing and utilizing new technologies like big data and digital technology. When used properly, these tools provide previously unheard of potential to spur innovation and advance our answer to the difficulties mentioned above. This review will go into greater detail about these terminologies later. It is critical to acknowledge that (a) their deployment is a crucial type of innovation in and of itself, and (b) that the usage of these technologies can stimulate more invention by increasing the efficacy of already-existing innovation systems (Sonka, 2016).

A distributed network of mutually anonymous nodes stores encrypted transactional data in an unchangeable record known as a blockchain (Mirabelli & Solina, 2020). Due to its many advantages for the agriculture industry, blockchain technology has drawn a lot of attention in recent years. This is because of its superior security, decentralized, unanimous, and unchangeable features (Rocha et al., 2021; Dasaklis et al., 2022). Reliable data must be available for expert systems and Internet of Things (IoT) devices to make well-informed decisions in smart and sustainable agriculture. In order for farmers, producers, agricultural communities, and suppliers to make well-informed decisions, the current e-agriculture platforms give them information (Schakel et al., 1997; Eitzinger et al., 2019). These technologies obtain trustworthy data from farmers and food producers, which greatly

increases production (Kamilaris et al., 2019). These methods do have several drawbacks, though, include being rigid, gathering insufficient data, and producing inaccurate information (Liu et al., 2021). Given the global distribution and multi-actor nature of the agricultural sector, which includes a wide range of players, including farmers, distributors, suppliers, wholesalers, retailers, financial institutions, and consumers, these systems are inappropriate. Numerous studies indicate that the best way to address the issues with these e-agriculture systems is through blockchain technology (Kouhizadeh et al., 2021). This is because of the essential features of blockchain, which include traceability, dependability, proof-of-ownership, proof-of-work and, encrypted and unchangeable data, which create reliable settings with few or no middlemen (Walter et al., 2017).

A digital revolution is taking place in the food business as a result of ongoing technological advancements (Dayioğlu & Turker, 2021). While big data analytics have given us a greater understanding of what customers want, automated production and smart manufacturing technologies have increased production efficiency. Furthermore, blockchain technology provides food safety security by improving the supply chain's traceability and transparency. Virtual reality and 3D-printed food are two examples of new technology that are simultaneously creating new opportunities for food innovation in the future. These technological developments offer a strong basis for the use of AI and big data, and this article will go into additional detail about these technologies' main functions and benefits in the food sector.

A collection of methods that boost agricultural productivity and lessen farming's environmental impact is referred to as "smart and sustainable agriculture" (Pedroso & Nakano, 2009). Numerous systems and technologies are involved, including information and communication technology (ICT) and the Internet of Things (IoT). In order to reach the consumers, a number of procedures are involved, including production on the farm, transformation, distribution, and trading (Pedroso & Nakano, 2009; National Science Foundation, 2012). From farm to fork, the food supply chain is a lengthy and intricate process with numerous middlemen and countless interactions. Financial transactions take place in the opposite direction of the flow of

goods and services. In today's agrifood supply networks, information is a valuable resource that goes both ways (Liu et al., 2021). But in a supply chain transaction, the asymmetric nature of knowledge and misinformation is a persistent problem. In addition, consumers are interested in knowing where the food they eat comes from. In addition to requiring a system to know the product information, they must check the sustainability of their products (Meadows et al., 1972). A sustainable system that regulates the conflict between the supply and demand of agrifood products, food provenance, and the removal of centralized authority is therefore under tremendous pressure on food suppliers.

2. BIG DATA APPLICATIONS

Despite being a relatively new concept, big data has been defined in a number of ways. For instance:

- "Big data" describes extensive, varied, intricate, long-term, and/or dispersed datasets produced by sensors, instruments, email, video, click streams, Internet transactions, and/or any other digital source that is currently or will be accessible in the future (National Science Foundation, 2012).
- According to (Chen et al., 2014), big data refers to datasets that are too large for typical IT and software/hardware solutions to view, collect, manage, and process in a reasonable amount of time.
- Big data refers to situations where the volume, acquisition speed, or variety of data representation make it impossible to conduct efficient analysis using conventional relational approaches or need the usage of substantial horizontal scaling for processing (Cooper & Mell, 2012).
- Big data is an information asset with large volume, high velocity, and high diversity that necessitates creative, affordable information processing methods for improved insight and decision-making (Gartner IT Glossary, 2012).

Consideration of the use of big data processes and techniques to agribusiness as a substantial chance for finance, ICT pack utilization, and gaining further relevance within the agriculture sector has been prevalent (Kamilaris et al., 2019;

Liu et al., 2021; Pedroso & Nakano, 2009). Food security concerns are lessened by the use of huge data in agriculture, which is not only harshly related to primary production but also plays a major role in improving the efficiency of the entire supply chain (Liu et al., 2021; Kouhizadeh et al., 2021). Discussions about the literary applications of big data are currently taking place mostly in America (Pedroso & Nakano, 2009), Canada (Dey & Shekhawat, 2021), Europe (Luzzani et al., 2021), and China (Saurabh & Dey, 2021). In-depth, advanced, game-changing business intelligence is centred on massive data, at a scale and pace than the previous technique. It is not suitable to duplicate and remove all of it into a warehouse at this time. Benchmarking, IoT-based sensor network deployment and analytics, prediction models, and using improved models to monitor crop failure risks and improve feed efficiency in livestock farming are some opportunities for Big Data application in agribusiness. Therefore, big data technology is expected to provide predictive insights into future farming outcomes, facilitate effective decision-making in real-time, and modernize business measures for quick, cutting-edge actions and revolutionary business models (Belotti et al., 2019) (Fig. 1).

People's needs for food have increased as a result of advancement and economic growth. Food nutrition and flavor have become more important to people (Schakel et al., 1997). The unique dietary requirements of every location and person are not satisfied by conventional food science and technology. The food business has seen dramatic developments in terms of supply chain optimization and food safety due to the emergence of artificial intelligence and big data analytics technology. The future of food safety lies in the internet, artificial intelligence, and big data. Big data plays a significant role in food safety (Marvin et al., 2017). Kazama et al. (2018) have utilized big data technology to create a neural network that depends on the ingredients and composition of the food and can translate recipes from one country to another. This completes the transformation of regional food styles and makes it simpler for people to sample cuisines from far-flung locations. People may now prioritize the nutritional value of the food on their plates and determine the nutritional value of food by analyzing recipes which is courtesy of a big data-based nutritional evaluation system (Kalra et al., 2020).

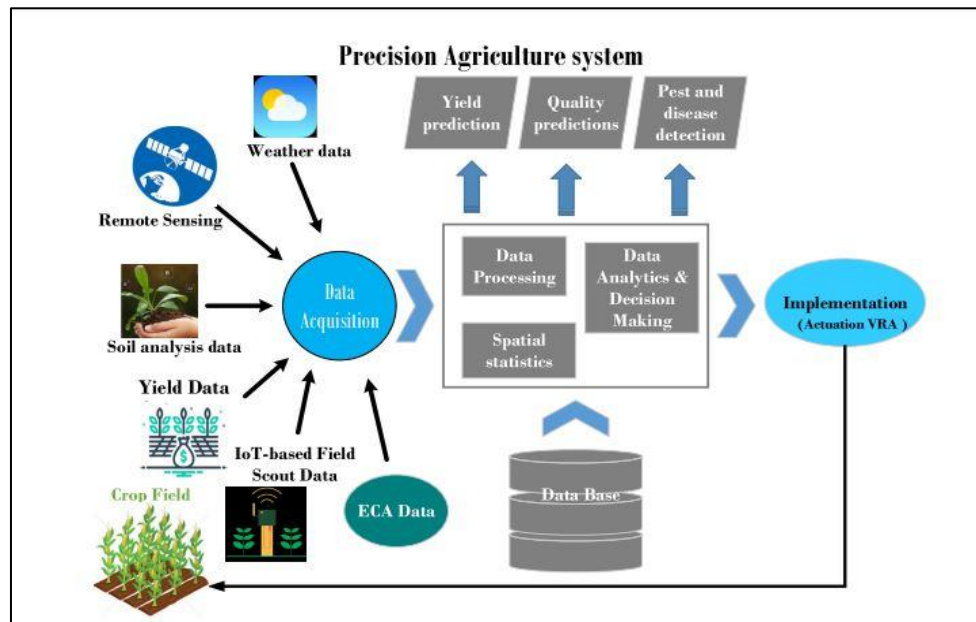


Fig 1. Big data-based precision agriculture system representation

3. THE BLOCKCHAIN TECHNOLOGY

An encrypted sequence of blocks is stored in a distributed record of transactions called a blockchain (Kalra et al., 2020; Mirabelli & Solina, 2020; Rocha et al., 2021). Information cannot be altered since the blockchain is designed to timestamp it. Blockchain technology is intended to reduce the impact of third parties, such as banks and brokers, on a transaction. Distributed ledger technology makes blockchain robust, safe, dependable, decentralized, and impervious to fraud. A decentralized shared database system is what blockchain is from a computational perspective. It is a public distributed ledger system from a technological standpoint, and it is a vast network of asset and data transactions without middlemen from a business standpoint (Mirabelli & Solina, 2020). Lack of trust, traceability, and transparency makes agriculture supply chains difficult. Furthermore, food epidemics in recent years have made customers' worries about food safety, quality, and provenance very important. "Even though various web-based e-agriculture systems are available and in use, these systems are not transparent and reliable as they are centralized, asymmetric, and incompatible" (Dayioğlu & Turker, 2021). Furthermore, consumers do not trust the current systems because there is no readily accessible, trustworthy information regarding the origins of food goods (Kalra et al., 2020). Thus, blockchain technology is a helpful instrument for

offering a practical resolution to the problems listed above.

Blockchain technology incorporates a number of current technologies for data collection, storage, and distribution (Mirabelli & Solina, 2020) which are (a) distributed and decentralized, with all network nodes dispersed throughout the network and no owner to oversee it, are the core features of blockchain technology, (b) tamper-proof and traceability, where the blockchain's algorithms guarantee the correctness and consistency of the data, (c) autonomous and peer-to-peer, in which the network's nodes are homologous because the blockchain network is fundamentally a peer-to-peer network; that is, the nodes are not connected to the network; and (d) automatic program execution, in which financial transactions, records, and data queries are carried out in accordance with pre-written programs known as smart contracts (Dasaklis et al., 2022). Anybody can participate and view transactions that take place on blockchain networks, which are typically accessible to all participants. These blockchains, which offer security and authentication, can be either public or private, or permissioned or permission-less. Blockchain makes it possible to read, verify, and store a sharable log of transactions in a series of blocks (Eitzinger et al., 2019). Blockchain systems operate in a distributed fashion in which P2P communications are used to connect the network to independent numerous agents, or participants (as nodes). Consequently, a

decentralized ecosystem devoid of a central authority is produced (Schakel et al., 1997; Ang & Seng, 2021).

E-agriculture systems have been researched for their ability to manage farms effectively and efficiently because they give users access to relevant data for well-informed decision-making, which boosts output (Serazetdinova et al., 2019; Marvin et al., 2017; Kazama et al., 2018). Nevertheless, human bias, a lack of trust and transparency in transactions, and asymmetric knowledge throughout supply chain operations are some of the drawbacks that have shown many e-agriculture systems to be ineffective. In comparison to e-agricultural systems, smart agriculture systems are more efficient and include cloud computing, mobile technologies, and sensor technologies like GPS and IoT enabled (Kamilaris et al., 2019; Kim et al., 2019). Intelligent systems can be used to manage livestock, water and waste management, soil conditions, field inspection, and remote monitoring. For these e-agriculture systems, the main issues have been data collection, data integrity, and data security. Additionally, it is simple for hackers to intercept and alter data from IoT-enabled farm management systems. The key characteristics of blockchain technology may be able to address the operational issues with smart e-agricultural systems (National Science Foundation, 2012; Jinesh, 2011).

4. UTILIZING BLOCKCHAIN TECHNOLOGY IN INTELLIGENT AND SUSTAINABLE FARMING

An immutable distributed ledger that is operated without a central authority is called blockchain (Liu et al., 2021). "A copy of every transaction is kept on every node (user) linked to the blockchain network, therefore the transactions stored there cannot be changed or deleted. The authenticity and transaction can therefore be confirmed in public. Blockchain is a distributed knowledge system that can serve as a dependable foundation for the long-term growth of smart agriculture applications. Its essential elements include proof-of-ownership, smart contracts, transparency, and consensus algorithms" (Kouhizadeh et al., 2021; Krizhevsky et al., 2012).

"In order to boost output, optimize input utilization, save labour costs, enhance market accessibility, and make timely decisions, farmers employ smart agriculture systems" (Ampatzidis &

Partel, 2019). For information collection, production management, record keeping, and equipment operation, they employ smart agriculture systems. 26% of farms utilize these systems for training and instruction, while over 80% use them on grain farms with GPS equipment (Walter et al., 2017; Ampatzidis & Partel, 2019). "The Bureau of Meteorology and other weather apps are examples of free internet software tools that some farmers use. Some of them make use of free software which is provided by private companies like AgWorld and GrainCorp" (Pedroso & Nakano, 2009). Significant gains in food supply chains and agricultural productivity have been demonstrated by the agricultural systems (Akella et al., 2021). "Precision farming, blockchain, IoT, and big data are all part of Agriculture 4.0, which is being considered by many researchers and organizations for all elements of sustainable agriculture. Precision agriculture is a farm management approach that focuses on measuring, tracking, and reacting to field metrics in order to quantify the output of crops or livestock for low-input, highly productive sustainable agriculture" (Pedroso & Nakano, 2009; Zhang, 2020; Cruz et al., 2017). Farmers may maintain sustainable agricultural practices and promote healthy crop growth by using these tools to visualize and evaluate data.

"Provenance, traceability, security, proof of work (PoW), and information asymmetry are all significant problems with current e-agriculture systems" (Mirabelli & Solina, 2020; Dey & Shekhawat, 2021; Luzzani et al., 2021; Saurabh & Dey, 2021; Song et al., 2021; Belotti et al., 2019). The implementation of blockchain technology is expected to benefit smart agriculture by adding new aspects. In order to identify the adoption facilitators and obstacles of blockchain technology for intelligent and sustainable agriculture, this report carries out extensive research. According to Luzzani et al. (2021), further research is needed on how to use blockchain technology to manage stakeholder trust in agriculture supply chains. "Decentralized applications are suggested as a remedy for problems in agriculture" (Saurabh & Dey, 2021). Blockchain technology and smart contracts may potentially help smart agriculture (Song et al., 2021). By 2030, IoT and blockchain-enabled smart agriculture applications will be accessible (Dey & Shekhawat, 2021). "They contend that the use of blockchain technology and the Internet of Things in agriculture is still in its infancy. The application of blockchain technology in

agriculture could streamline the financing and credit-checking procedures. The use of blockchain technology and big data to argue that a safe platform for sharing data is required" (Belotti et al., 2019). Significant study has been conducted on integrating blockchain technology into the agriculture sector, according to the literature review. However, technical obstacles have severely hindered the adoption of sustainable farming techniques and integrated smart agriculture systems.

Additionally, research indicates that the obstacles to embracing any new technology are generally generic, and the usefulness of blockchain for sustainable practices and smart agriculture is not sufficiently highlighted.

"Issues with trust and dependability arise in the farm-to-consumer scenario due to intricate procedures, knowledge gaps, and conflicting information between the participants" (Benyam et al., 2021; Ronaghi, 2021). The current study provides proof of blockchain technology's potential for agricultural supply chains and suggests solutions including traceability systems, secure payments, and provenance. "The studies also look for industry-level best practices and worldwide standards, as well as a few security vulnerabilities. In determining the degree of adoption, obstacles, and facilitators of blockchain technology for intelligent and sustainable agriculture, we find that there aren't many unique studies" (Demestichas et al., 2020; Jordan, 2004; Wainwright & Jordan, 2008; Lafferty et al., 2001). So that the feasibility of blockchain technology is fully understood and more insights can be gained, this review will fill in the gaps by analyzing adoption levels, enablers, and impediments. Significant adoption challenges and facilitators exist for blockchain technology adoption for intelligent and sustainable agriculture, according to a thorough analysis of the literature.

5. IMPLICATIONS OF DIGITAL TECHNOLOGY IN PRECISION AGRICULTURE

Collection and analyzing vast amounts of data generated by wireless sensor networks and IoT networks, including digital photos and additional data from satellites, UAVs, and data fusion. There are obstacles to the successful implementation of smart farming with the data now available. Potential ways to derive insights from said data include emerging technologies

related to data mining and artificial intelligence (Wytock & Kolter, 2013; Vatsavia et al., 2012; Kussul et al., 2017). These methods can help quickly and accurately identify trends, find hidden models, and explore larger and more unpredictable amounts of data. For a variety of reasons examined below, farming has not placed enough weight on these tactics' capacity for large data analysis. To sum up, the majority of the open frameworks that are currently available were created as a result of recent efforts, and their problem is still widely accepted. It is possible that many of them are still developing and haven't reached their full potential. "Large industrial farms (like Monsanto) that use big data in their decision-making processes and have the infrastructure to access data, resources, and—most importantly—finance are well-suited for the majority of these big data applications" (Ghatrehsamani et al., 2018).

In the poor world, not much is done on tiny farms. Big data has the potential to help non-industrial farms, but in order to achieve these benefits, moral and ethical concerns about funding, availability, and cost must be resolved.

6. CONCLUSION

The increasing availability of information due to ICT advancements seems to hold promise for advancing advances in essential decision-making by increasing model precision and generalization capacity. It is also expected that learning from the vast amount of data produced by precision agriculture techniques would lead to significant opportunities and revolutionary viewpoints for precision farming. Due to the development of big data, conventional learning techniques are neither inherently effective nor scalable enough to handle vast amounts of multi-dimensional, spatiotemporal, and heterogeneous data. Cutting-edge machine learning techniques like CNN and big data analysis approaches offer improved performance, vigour, adaptability, and precision. Beyond precision farming, automation and the use of AI, drones, IoT, robotics, and big data are expected to play a major role in a number of agricultural domains. High-performance data-driven scalable learning methods can transform conventional farm management into AI systems, enhancing real-time decision-making abilities and automating various agricultural tasks. The emergence of advanced machine learning and data mining

fields, alongside readily available datasets and strategic frameworks, is essential for addressing sustainability, efficiency, climate change, and food security challenges in agricultural production.

Blockchain technology was first created as the technology underlying cryptocurrencies and internet-native money. It now has a wide range of uses, including providing digital infrastructure for the next generation of global supply chains and agricultural trade. Blockchain technology has an impact on any business that uses digital data, and the agricultural industry is no different. By integrating blockchain technology, the agricultural industry would benefit from the digitization of trading and operational aspects, which could subsequently be transferred to blockchain technology as digital infrastructure. Additionally, this benefit is realized across the value chain, and the financial viability of individual farms will depend on the sector's coordination of the adoption of new technology. Some significant problems encountered in early-stage technology development include the relative lack of knowledge with the technology among parties and the need to coordinate adoptions. The new technology could be difficult, complex, and disruptive. If it is explained, it might have a significant positive impact on agricultural farmers by raising the price margin. The potential issues it resolves are enormous. It might be compelling evidence to recommend blockchain technology investment in the agriculture industry.

A data-driven system is advantageous to all parties involved in the agriculture industry, including farmers, consumers, financial institutions, the food processing sector, and many more. Despite the fact that its full potential for creating value has yet to be fully realized, it has already started to bring about massive changes in the agricultural sector. The emergence of improved and healthier products resulting from new plant genome sequencing methods, precision agriculture approaches that support logical decision-making, and the application of IoT sensor devices and analytical techniques that help reduce food waste and food-borne diseases are just a few of the numerous benefits provided by artificial intelligence and big data.

DISCLAIMER (ARTIFICIAL INTELLIGENCE)

Author(s) hereby declare that NO generative AI technologies such as Large Language Models

(ChatGPT, COPILOT, etc) and text-to-image generators have been used during writing or editing of this manuscript.

COMPETING INTERESTS

Authors have declared that they have no known competing financial interests or non-financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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